Steven Peng

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EDUCATION

University of Pennsylvania — **Philadelphia, PA**
B.A. in Physics & Computer Science — *Expected Class of 2030*

Nanjing Foreign Language School — **Nanjing, CN**
High School Diploma — *Class of 2026*

SAT: 1550 | TOEFL: 117 | AP 5s: Physics C (E&M, Mechanics), Calculus BC, Statistics

PROFESSIONAL EXPERIENCE

Horizon Robotics Inc. — **Beijing, CN**
Robotics Lab Intern, Physics & AI — *Summer 2026*

- Developed Rally3D and Hand2Body for a humanoid table-tennis pipeline on the 29-DoF Unitree G1, converting human video into whole-body motion through GMR retargeting and HoloMotion tracking at 50 Hz with 500 Hz motor PD; credited author on HoloMotion v2.0.
- Sole-authored Rally3D, a resumable pipeline converting monocular WTT/ITTF footage into synchronized metric 3D ball trajectories, player poses, and table/camera geometry using RTMPose, MotionBERT, BlurBall, camera calibration, and physics-based lifting; reached up to 92% 2D ball coverage on clean rallies and extracted 98 rally clips from one full match.
- Sole-authored Hand2Body, a causal streaming DiT diffusion model mapping 1–2 wrist trajectories to 135-D whole-body SMPL motion; trained the table-tennis model on 7,753 sequences / ~4.14M frames and achieved 20.7 mm MPJPE and 8.5 mm / 0.72° wrist tracking, outperforming a transformer-regressor baseline on every metric.
- Implemented streaming inference at ~1.6 ms/frame with ~130 ms latency, FK hand-consistency and anti-skate losses, and an 18-file pytest suite; integrated generated SMPL-X motion through GMR → HoloMotion, MuJoCo QC/sim2sim evaluation, and the documented ROS2 on-robot deployment path.

Robotics Lab Intern, Physics & AI — *Summer 2025*

- Built a Stanford ToddlerBot with Dynamixel actuators and Jetson Orin compute; trained MJX/Brax RL locomotion policies with gin-configured environments and Weights & Biases and TensorBoard experiment tracking.
- Deployed policies sim-to-real through hardware calibration, domain randomization, and zero-shot transfer; ran a Mac-development / Jetson-execution workflow over SSH using URDF/MJCF robot descriptions.

IYPT Team China — **Lund, Sweden**
Team Captain — *Sep 2023 – Present*

- Selected 1 of 5 national team members through 5 rounds and elected Captain; led Team China to Gold Medal and Global 2nd at IYPT 2025, achieving the team's all-time high score.
- Designed experiments and built quantitative models for 15+ open-ended physics problems; led real-time tactical decisions in competitive scientific debates across international rounds.

PROJECTS & RESEARCH

Acoustic Flame Suppression Research — **Beijing, CN**
Researcher — *Sep 2025*

- Studied acoustic flame extinction with 400K-fps schlieren photography on an unsolved IYPT problem.
- Proposed vortex-induced O_2 depletion as the primary mechanism, achieving state-of-the-art theory-experiment matching.

Quantum Entangled Light Beam Generation, Nanjing University — **Nanjing, CN**
Researcher, Prof. Xiaosong Ma — *Oct 2024*

- Generated entangled photon pairs via SPDC under Prof. Xiaosong Ma; measured 16 polarization combinations to test the CHSH inequality, achieving $S = 2.487$ vs. the classical limit of 2.

S.-T. Yau Science Award Project & Non-Invasive Acoustic Counting — **Beijing, CN**
Researcher — *Oct 2024*

- Built a CNN-based non-invasive counting method from acoustic signatures, earning an invention patent and publication in the Cambridge Journal for Junior Scientists; modeled acoustic energy transfer in ball collisions and validated results via simulation, winning Gold at the 2024 S.-T. Yau High School Science Award (Physics) Global Finals as Team Captain.

HONORS

2024: USAMO Qualifier | Duke Math Meet China Regional: 1st Individual & Team, Captain
2024: HMMT Geometry 13th/651 | Physics Bowl National Gold | 2023: AMC 12B Perfect Score

SKILLS

Languages: Python, C, Assembly, JavaScript, MATLAB, SQL
Frameworks/Tools: PyTorch, Linux/Unix, MuJoCo, CUDA, ROS2, Isaac Sim, Weights & Biases, TensorBoard, Postgres